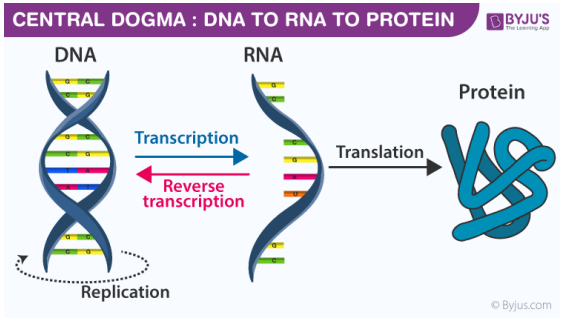
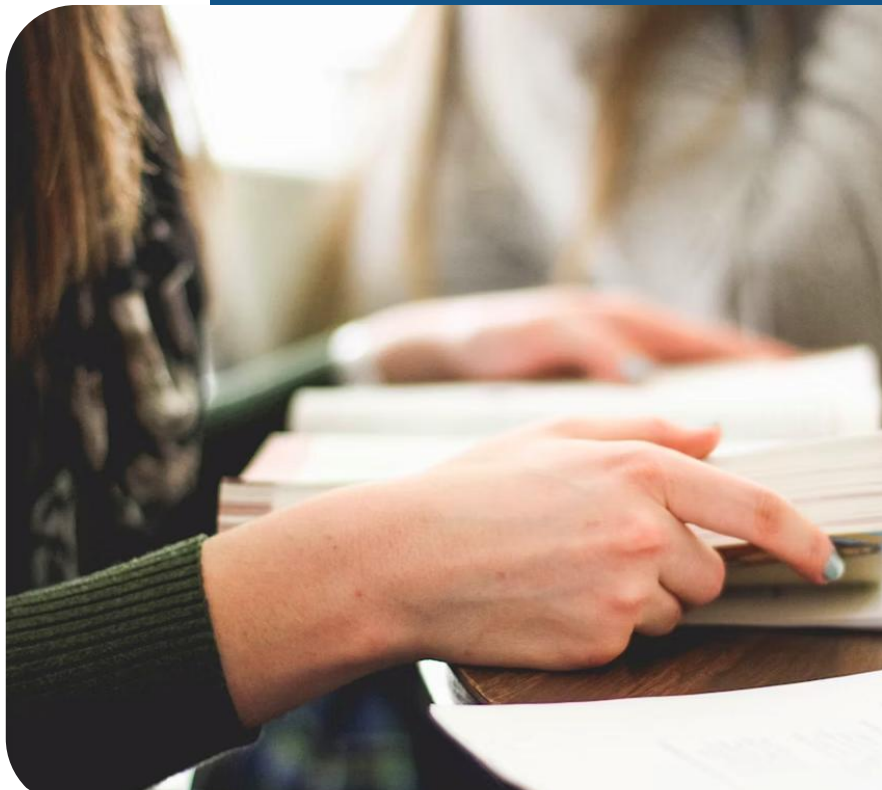
Central Dogma of Biology



Translation



TRIPLET / CODON

T A G C A G A T G

DNA

'SERINE'

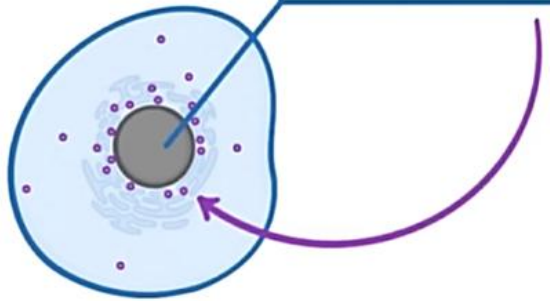
A G U C C A U C G

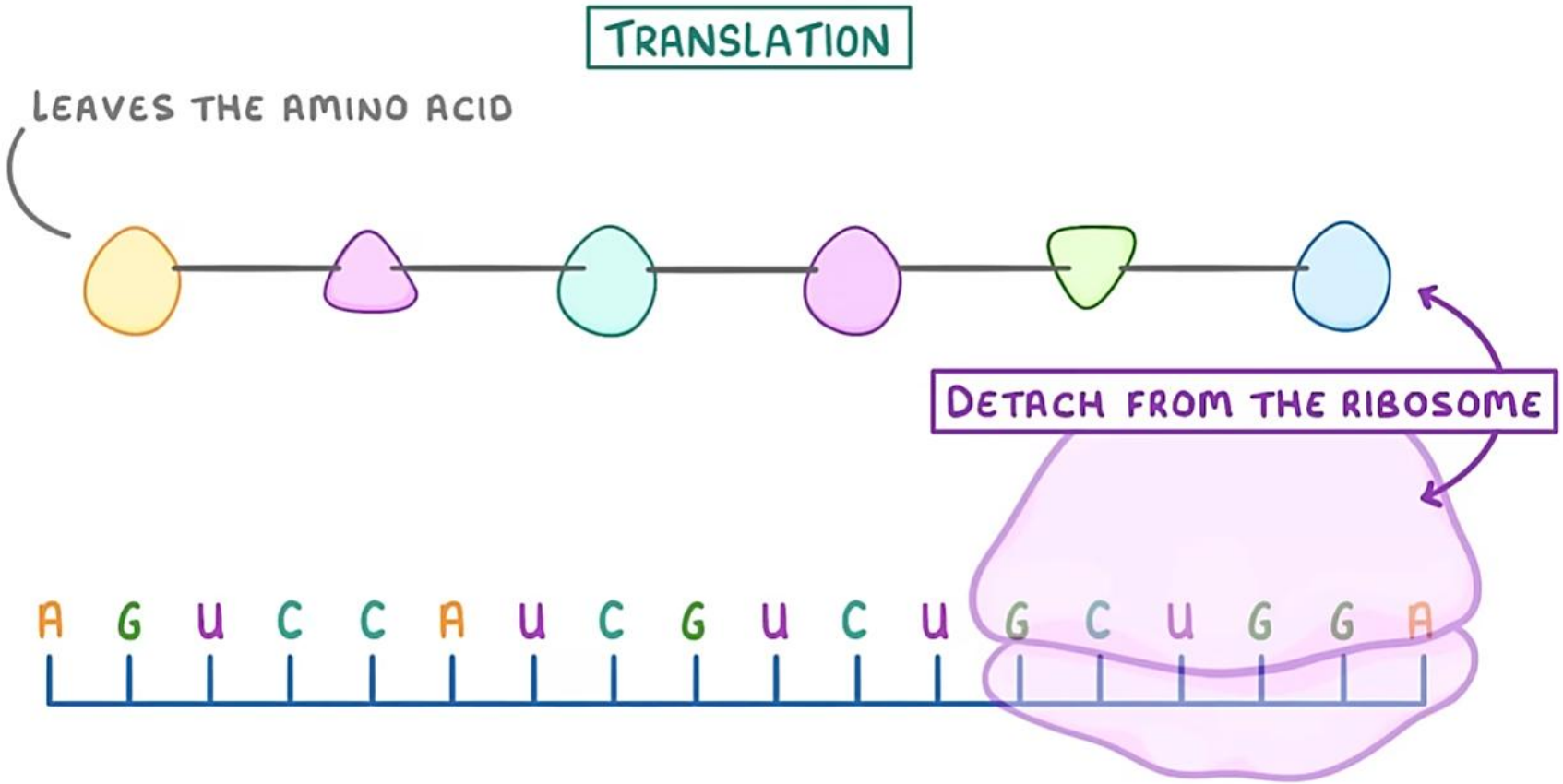
mRNA

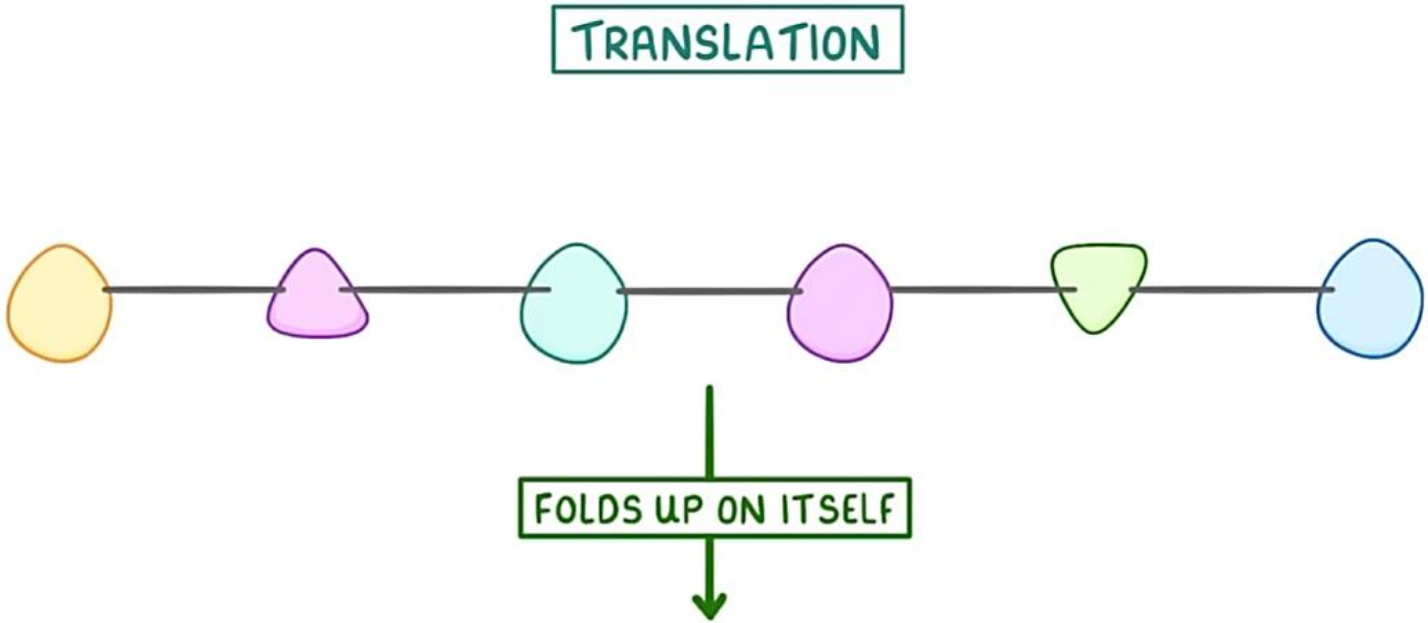
TRANSLATION

mRNA

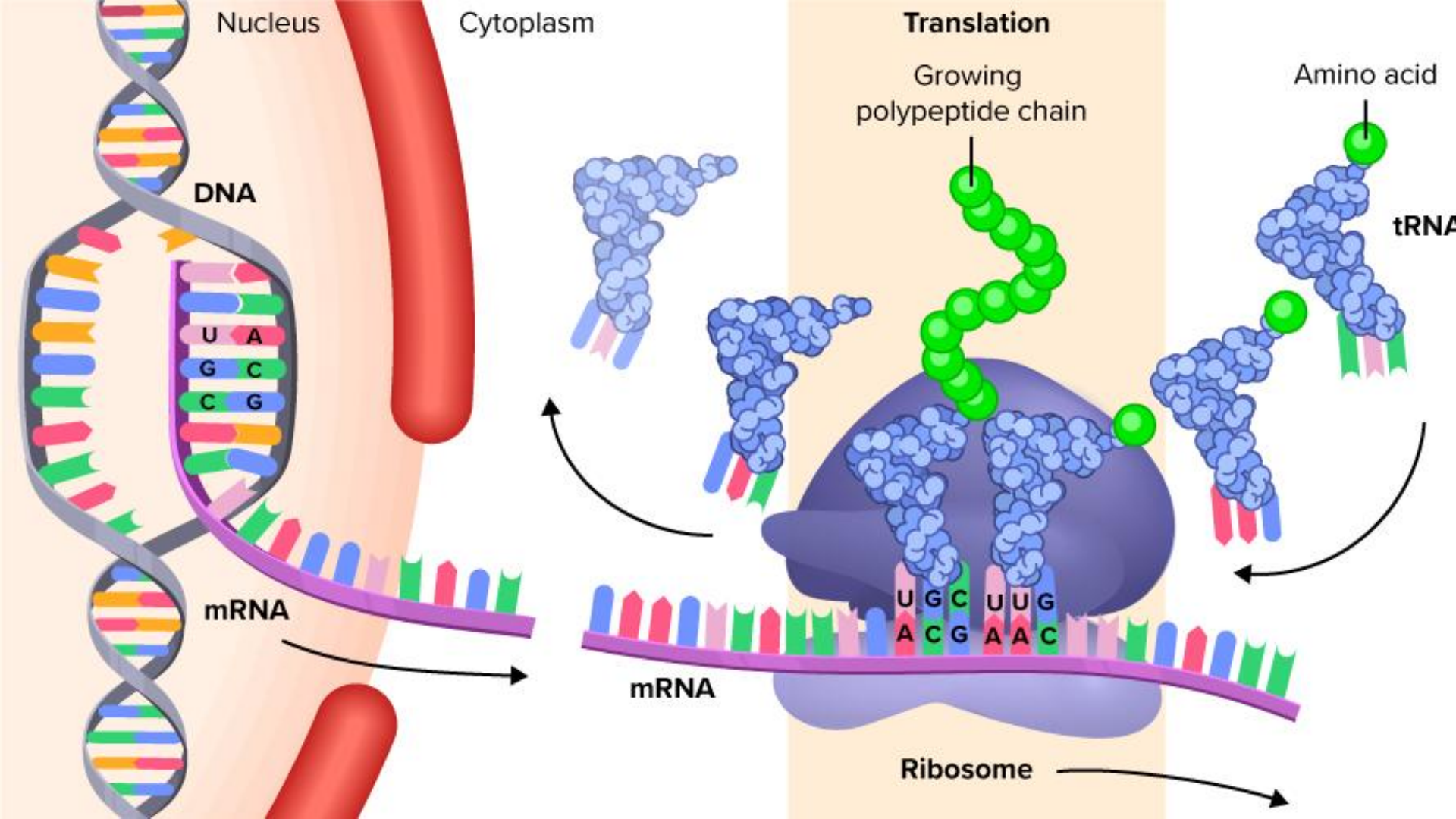
A G u C C A u C G u C u G C u G G A

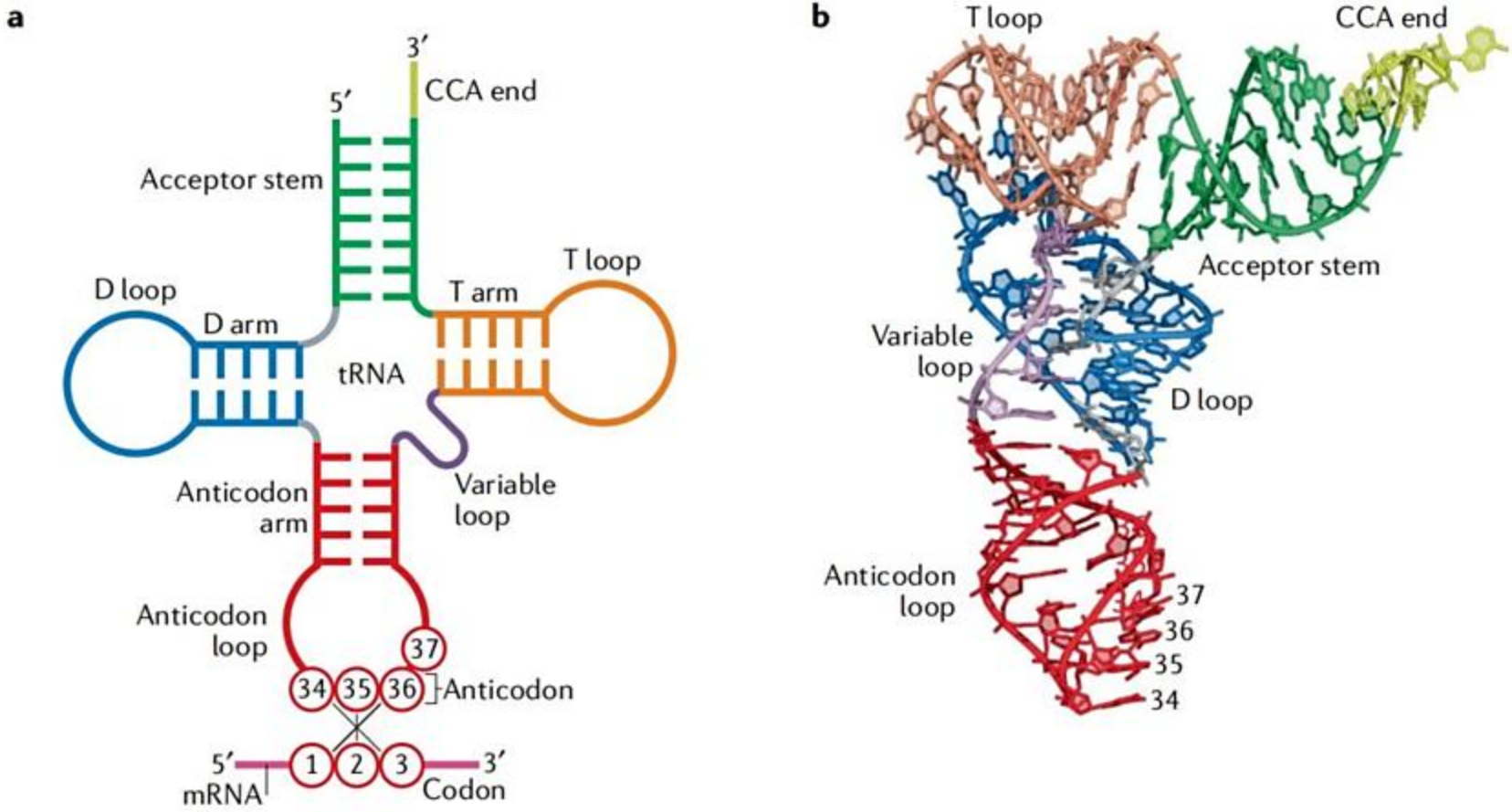



TRANSLATION
LEAVES THE AMINO ACID
DETACH FROM THE RIBOSOME
A G U C C A U C G U C U G C U G G A



The human body can synthesize **over 100,000 different types of proteins**. These proteins are constructed from only 20 standard amino acids, 11 of which the body can produce itself, while 9 essential amino acids must be obtained from diet. These proteins are encoded by approximately 20,000-25,000 genes.


Nucleus
Cytoplasm
Translation
Growing polypeptide chain
Amino acid
DNA
tRNA
U A
G C
C G
mRNA
U G C U U G
A C G A A C
mRNA
Ribosome


a
3′
CCA end
5′
Acceptor stem
T loop
D loop
T arm
D arm
tRNA
Variable
loop
Anticodon
arm
Anticodon
loop
37
34
35
36
Anticodon
5′
1
2
3
3′
mRNA
Codon
b
T loop
CCA end
Acceptor stem
Variable
loop
D loop
Anticodon
loop
37
36
35
34

First Base	Second Base: U		Second Base: C		Second Base: A		Second Base: G		Third Base
U	UUU	phe	UCU	ser	UAU	tyr	UGU	cys	U
	UUC		UCC		UAC		UGC		C
	UUA	leu	UCA		UAA	STOP	UGA	STOP	A
	UUG		UCG		UAG		UGG	trp	G
C	CUU	leu	CCU	pro	CAU	his	CGU	arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	ile	ACU	thr	AAU	asn	AGU	ser	U
	AUC		ACC		AAC		AGC		C
	AUA		ACA		AAA	lys	AGA	arg	A
	AUG	met	ACG		AAG		AGG		G
G	GUU	val	GCU	ala	GAU	asp	GGU	gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	glu	GGA		A
	GUG		GCG		GAG		GGG		G

Quiz

DNA template strand (3′ → 5′):
3′ – TAC GAA TTT CGC ATG TAA CCT ATT– 5′

Create the mRNA and amino acid sequence!

Remember that:

- DNA template strand is read 3′ → 5′
- mRNA is synthesized 5′ → 3′
- RNA uses uracil (U) instead of thymine (T)
- Translation begins at the start codon (AUG) and ends at a stop codon (UAA, UAG, or UGA)

Quiz

Given **DNA template strand** (3′ → 5′):
3′ – TAC GAA TTT CGC ATG TAA CCT ATT– 5′

Using base-pairing rules (A↔U, T↔A, C↔G, G↔C):
mRNA (5′ → 3′):
5′ – AUG CUU AAA GCG UAC AUU GGA UAA– 3′

mRNA Codon	Amino Acid
AUG	Methionine (Met)
CUU	Leucine (Leu)
AAA	Lysine (Lys)
GCG	Alanine (Ala)
UAC	Tyrosine (Tyr)
AUU	Isoleucine (Ile)
GGA	Glycine (Gly)
UAA	Stop

Met – Leu – Lys – Ala – Tyr – Ile – Gly

THANK YOU!